

A Consistency-Gated Quaternion Error-State Kalman Filter for Multi-Sensor Inertial Navigation

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Abstract

We describe the design, implementation, and verification of an error-state Kalman filter (ESKF) that fuses a strapdown inertial measurement unit (IMU) with GPS, a barometric altimeter, a magnetometer, a downward laser altimeter (LiDAR), ultra-wideband (UWB) radio ranging, and optical flow to estimate the pose of an aerial vehicle. The filter follows the multiplicative, local error-state formulation of Solà [2] and Trawny and Roumeliotis [3]: a nominal state carries the full estimate and is integrated directly from the IMU, while a minimal 15-dimensional error state, reset to zero after every correction, carries the covariance. Each measurement model is given with its exact error-state Jacobian, and every Jacobian is validated against a finite difference. The central claim is not merely that the filter tracks well but that its *reported uncertainty is correct*: we verify this with the normalized estimation error squared (NEES) consistency test of Bar-Shalom et al. [5], run as a Monte-Carlo gate over many simulated flights. On the full seven-sensor problem the filter is statistically consistent (position NEES 2.84 against an expected 3.0; full-state NEES 15.6 against 15.0), while remaining faithful to inertial-navigation error theory: with all aiding removed, the position error grows without bound, and it is recovered to sub-metre accuracy when a single aiding modality, including UWB ranging alone, is restored.

1 Introduction

Aided inertial navigation estimates a vehicle’s position, velocity, and orientation by integrating high-rate inertial measurements and correcting the accumulated drift with lower-rate aiding sensors [6, 7]. Orientation makes the problem subtle: a unit quaternion has four components but only three degrees of freedom, so a filter that estimates it directly must either enforce the unit-norm constraint or carry a covariance for a quantity that is not free to move in four directions. The *error-state* (or indirect, or multiplicative) Kalman filter resolves this by estimating a minimal three-parameter rotation error in the tangent space of the current attitude estimate [4, 2]. This paper documents such a filter, its six measurement models, and—most importantly—the statistical test that proves its covariance honest.

The contribution is one of rigor and reproducibility rather than novelty of formulation: the filter is a faithful implementation of the standard local-error ESKF, but each Jacobian is finite-difference-checked, the whole estimation loop is gated on NEES consistency, and the entire system—core, simulator, and a real-time three-dimensional visualization—runs in the browser via WebAssembly and WebGL2. Source and an interactive demonstration are available at <https://github.com/Zarathos94/ekf-sim> and <https://irfansaric.com/showcase/ekf-sim>.

2 State Representation

The **nominal state** collects position $\mathbf{p} \in \mathbb{R}^3$, velocity $\mathbf{v} \in \mathbb{R}^3$, orientation as a Hamilton unit quaternion $\mathbf{q} \in \mathbb{S}^3$ (body to world), and the accelerometer and gyroscope biases $\mathbf{a}_b, \boldsymbol{\omega}_b \in \mathbb{R}^3$:

$$\mathbf{x} = (\mathbf{p}, \mathbf{v}, \mathbf{q}, \mathbf{a}_b, \boldsymbol{\omega}_b). \quad (1)$$

The **error state** is the 15-vector

$$\delta\mathbf{x} = (\delta\mathbf{p}, \delta\mathbf{v}, \delta\boldsymbol{\theta}, \delta\mathbf{a}_b, \delta\boldsymbol{\omega}_b) \in \mathbb{R}^{15}, \quad (2)$$

where the rotation error $\delta\boldsymbol{\theta}$ is defined *locally* (in the body frame) through the multiplicative relation

$$\mathbf{q}_{\text{true}} = \mathbf{q} \otimes \exp\left(\frac{1}{2}\delta\boldsymbol{\theta}\right), \quad (3)$$

with $\exp(\cdot)$ the quaternion exponential map and $\otimes$ the Hamilton product. The nominal state holds the full, unconstrained estimate; the error state, kept near zero, holds a covariance $\mathbf{P} \in \mathbb{R}^{15 \times 15}$ whose orientation block is an ordinary 3×3 matrix rather than a constrained 4×4 one. We write $\mathbf{R} = \mathbf{R}(\mathbf{q})$ for the rotation matrix and $[\cdot]_{\times} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ for the skew-symmetric “cross-product” operator, so that $[\mathbf{a}]_{\times} \mathbf{v} = \mathbf{a} \times \mathbf{v}$.

3 Propagation

At each IMU sample (rate ≈ 200 Hz, interval Δt) with accelerometer reading $\mathbf{a}_m$ and gyroscope reading $\boldsymbol{\omega}_m$, define the bias-corrected specific force and body rate

$$\mathbf{a} = \mathbf{a}_m - \mathbf{a}_b, \quad \boldsymbol{\omega} = \boldsymbol{\omega}_m - \boldsymbol{\omega}_b. \quad (4)$$

Nominal integration. With gravity $\mathbf{g} = (0, 0, -9.806\,65 \text{ m s}^{-2})$,

$$\mathbf{p} \leftarrow \mathbf{p} + \mathbf{v} \Delta t + \frac{1}{2}(\mathbf{R}\mathbf{a} + \mathbf{g}) \Delta t^2, \quad (5)$$

$$\mathbf{v} \leftarrow \mathbf{v} + (\mathbf{R}\mathbf{a} + \mathbf{g}) \Delta t, \quad (6)$$

$$\mathbf{q} \leftarrow \mathbf{q} \otimes \exp\left(\frac{1}{2}\boldsymbol{\omega} \Delta t\right), \quad (7)$$

with the quaternion renormalized after each step; the biases are constant under nominal integration.

Error-state transition. Linearizing the local error dynamics gives $\delta\mathbf{x} \leftarrow \mathbf{F} \delta\mathbf{x}$ with the block matrix

$$\mathbf{F} = \begin{bmatrix} \mathbf{I} & \mathbf{I} \Delta t & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & -\mathbf{R}[\mathbf{a}]_{\times} \Delta t & -\mathbf{R} \Delta t & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \exp(-[\boldsymbol{\omega}]_{\times} \Delta t) & \mathbf{0} & -\mathbf{I} \Delta t \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix}, \quad (8)$$

in which the velocity-error coupling $-\mathbf{R}[\mathbf{a}]_{\times}$ is the first-order effect of an orientation error on the rotated specific force, and the orientation block $\exp(-[\boldsymbol{\omega}]_{\times} \Delta t)$ is the exact rotation integral (evaluated via Rodrigues’ formula) rather than its first-order truncation [2].

Process noise. The covariance propagates as $\mathbf{P} \leftarrow \mathbf{F} \mathbf{P} \mathbf{F}^{\top} + \mathbf{Q}$ with the block-diagonal

$$\mathbf{Q} = \text{diag}(\mathbf{0}, \sigma_a^2 \Delta t^2 \mathbf{I}, \sigma_g^2 \Delta t^2 \mathbf{I}, \sigma_{a_b}^2 \Delta t \mathbf{I}, \sigma_{\omega_b}^2 \Delta t \mathbf{I}). \quad (9)$$

The IMU white-noise terms scale as Δt^2 : a per-sample noise of standard deviation σ perturbs the integrated velocity or orientation by $\sigma \Delta t$ over the step, injecting a variance $\sigma^2 \Delta t^2$ (this is also the only dimensionally consistent choice— $\sigma^2 \Delta t$ would carry units of m^2/s^3 , not $(\text{m/s})^2$). The bias random walks scale as Δt . A mis-scaled $\mathbf{Q}$ is invisible in the RMSE but glaring in the NEES of Section 6.

4 Measurement Updates

For a measurement $\mathbf{z}$ with model $\mathbf{z} = h(\mathbf{x}) + \boldsymbol{\nu}$, $\boldsymbol{\nu} \sim \mathcal{N}(0, \mathbf{R})$, the Jacobian $\mathbf{H} = \partial h / \partial \delta \mathbf{x}$ is evaluated at the nominal state and the correction is the standard sequence

$$\mathbf{S} = \mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R}, \quad \mathbf{K} = \mathbf{P}\mathbf{H}^\top \mathbf{S}^{-1}, \quad \delta \mathbf{x} = \mathbf{K}(\mathbf{z} - h(\mathbf{x})), \quad (10)$$

followed by injection (Section 5) and the Joseph-form covariance update

$$\mathbf{P} \leftarrow (\mathbf{I} - \mathbf{K}\mathbf{H}) \mathbf{P} (\mathbf{I} - \mathbf{K}\mathbf{H})^\top + \mathbf{K}\mathbf{R}\mathbf{K}^\top, \quad (11)$$

which preserves symmetric positive-definiteness under finite precision where the shorter $(\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}$ form drifts. The six models, each with the non-zero blocks of $\mathbf{H}$, are:

- **GPS** (position): $h(\mathbf{x}) = \mathbf{p}$, so $\partial h / \partial \delta \mathbf{p} = \mathbf{I}_3$.
- **Barometer** (altitude): $h(\mathbf{x}) = \mathbf{e}_3^\top \mathbf{p}$, so $\partial h / \partial \delta \mathbf{p} = \mathbf{e}_3^\top$.
- **Magnetometer**: the world reference field $\mathbf{m}$ resolved in the body frame, $h(\mathbf{x}) = \mathbf{R}^\top \mathbf{m}$. Using (3), $\partial h / \partial \delta \boldsymbol{\theta} = [\mathbf{R}^\top \mathbf{m}]_\times$, giving heading observability from the full vector rather than a yaw pseudo-measurement.
- **LiDAR altimeter**: a downward laser range to flat ground $z = 0$ along the tilted body axis, $h(\mathbf{x}) = p_z / R_{33}$, where $R_{33} = \mathbf{e}_3^\top \mathbf{R} \mathbf{e}_3$. Both $\partial h / \partial p_z = 1 / R_{33}$ and $\partial h / \partial \delta \boldsymbol{\theta} = -\frac{p_z}{R_{33}^2} [-R_{32}, R_{31}, 0]$ are non-zero, so a single beam senses height *and* attitude.
- **UWB radio ranging**: scalar range to a fixed beacon $\mathbf{b}_k$ at a known position, $h(\mathbf{x}) = \|\mathbf{p} - \mathbf{b}_k\|$, with $\partial h / \partial \delta \mathbf{p} = \hat{\mathbf{u}}^\top$, the unit line of sight. Being position-only, a few beacons localize the vehicle with no GPS at all [8].
- **Optical flow**: the body-frame horizontal velocity $h(\mathbf{x}) = \Pi \mathbf{R}^\top \mathbf{v}$, with Π selecting the first two components, so $\partial h / \partial \delta \mathbf{v} = \Pi \mathbf{R}^\top$ and $\partial h / \partial \delta \boldsymbol{\theta} = \Pi [\mathbf{R}^\top \mathbf{v}]_\times$; it observes velocity and attitude [9].

Every Jacobian above is checked in the test suite against a central finite difference of its measurement function across all 15 error directions, agreeing to 10^{-4} .

5 Injection and Reset

The error-state correction is folded into the nominal state and the error reset to zero:

$$\mathbf{p} \leftarrow \mathbf{p} + \delta \mathbf{p}, \quad \mathbf{v} \leftarrow \mathbf{v} + \delta \mathbf{v}, \quad \mathbf{q} \leftarrow \mathbf{q} \otimes \exp(\tfrac{1}{2} \delta \boldsymbol{\theta}), \quad \mathbf{a}_b \leftarrow \mathbf{a}_b + \delta \mathbf{a}_b, \quad \boldsymbol{\omega}_b \leftarrow \boldsymbol{\omega}_b + \delta \boldsymbol{\omega}_b. \quad (12)$$

Because the linearization point of the orientation error moves onto the new nominal quaternion, the covariance is transformed by the reset Jacobian

$$\mathbf{P} \leftarrow \mathbf{G} \mathbf{P} \mathbf{G}^\top, \quad \mathbf{G} = \text{diag}(\mathbf{I}, \mathbf{I}, \mathbf{I} - \tfrac{1}{2} [\delta \boldsymbol{\theta}]_\times, \mathbf{I}, \mathbf{I}). \quad (13)$$

The term $\mathbf{I} - \frac{1}{2} [\delta \boldsymbol{\theta}]_\times$ is close to identity and is frequently omitted; retaining it is one of the small details the consistency test rewards.

6 Consistency: The NEES Gate

An estimator that tracks well but reports the wrong uncertainty is not merely inelegant—it will mislead whatever consumes its covariance. The decisive test is *statistical consistency*: the normalized estimation error squared [5]

$$\varepsilon = \mathbf{e}^\top \mathbf{P}^{-1} \mathbf{e}, \quad (14)$$

where $\mathbf{e}$ is the estimation error expressed in the same 15-dimensional tangent space that $\mathbf{P}$ describes (position and velocity differences, the body-frame logarithmic map for orientation, and bias differences). For a consistent filter, $\mathbb{E}[\varepsilon] = \dim$. Averaged over N independent Monte-Carlo runs, $N\bar{\varepsilon} \sim \chi^2(N \cdot \dim)$, which yields the two-sided 95% acceptance interval

$$\bar{\varepsilon} \in \left[\frac{\chi_{0.025}^2(N \cdot \dim)}{N}, \frac{\chi_{0.975}^2(N \cdot \dim)}{N} \right]. \quad (15)$$

A filter that is overconfident (covariance too small) lands above the band; one that is conservative lands below. The runs are seeded from a realistic initial error drawn from the initial covariance, so the test scores the filter and not an omniscient initialization.

7 Simulation Environment

Ground truth is an analytic banking helical trajectory—a horizontal orbit of radius 40 m and period 30 s with a superimposed vertical oscillation—so position, velocity, and acceleration are exact and no integration error contaminates the reference. Orientation follows a coordinated turn (nose along velocity, rolled into the turn by the bank angle that balances lateral acceleration against gravity), and the body angular rate is obtained by central-differencing the quaternion path. From this truth a strapdown IMU (specific force and body rate, with bias random walk and white noise), and lower-rate GPS, barometer, magnetometer, LiDAR, UWB, and optical-flow measurements are synthesized. Every noise level, bias-walk density, and per-sensor availability is a parameter, so degradation and recovery can be driven interactively.

8 Results

Table 1 reports the Monte-Carlo consistency check over 40 flights of 30 s with all seven sensors active. Both the position (3-DOF) and full-state (15-DOF) NEES fall inside the 95% band of (15): the filter’s covariance matches its error. The time-averaged RMSE over the same runs is 0.080 m in position, 0.042 m s^{−1} in velocity, and 0.222° in attitude.

Quantity	NEES mean	Expected	95% band
Position (3-DOF)	2.84	3.0	[2.29, 3.81]
Full state (15-DOF)	15.64	15.0	[13.35, 16.74]

Table 1: NEES consistency over 40 Monte-Carlo flights. Both statistics lie inside the band, i.e. the reported uncertainty is neither over- nor under-confident.

Table 2 shows steady-state RMSE as sensors are removed or degraded. Two rows make the inertial-navigation physics visible. With *all* aiding removed, the filter dead-reckons on the IMU alone and the position error grows without bound, reaching hundreds of metres over 30 s—exactly the quadratic error growth that accelerometer and gyroscope bias predict for an unaided strapdown system [7]. Yet a single position-fixing modality restores sub-metre accuracy: with GPS denied but UWB ranging enabled, the error collapses from 16.6 m to 0.16 m, confirming that the radio-ranging model and its Jacobian are correct and that the filter degrades and recovers gracefully.

Scenario	Position (m)	Velocity (m/s)	Attitude (°)
Nominal (GPS + baro + mag)	0.47	0.24	0.99
IMU only (dead reckoning)	295	25.4	8.3
GPS dropout	16.6	1.87	2.05
GPS denied + UWB ranging	0.16	0.10	0.61
Noisy IMU (5×)	0.48	0.26	1.06
Bias drift (10×)	0.52	0.33	1.53

Table 2: Steady-state RMSE by scenario (mean of 12 flights of 30s each). Dead reckoning diverges; any single position fix, including UWB alone, recovers it.

9 Conclusion

We have presented a quaternion error-state Kalman filter fusing seven sensors for inertial navigation, in the standard local-error formulation, with each measurement Jacobian derived and finite-difference-verified. The emphasis throughout is that correctness is *demonstrated*, not asserted: the full estimation loop passes a Monte-Carlo NEES consistency gate, and the filter reproduces the expected inertial-navigation behavior—unbounded dead-reckoning drift, and graceful recovery from any single position aid, including GPS-denied UWB ranging. The complete system, from the dependency-free numerical core to a real-time 3D visualization, is open source and runs in the browser.

References

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